Year 12 Further Maths - Vectors

Similar questions - Dot product, cross product and angles

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) Find a vector which is perpendicular to both $\begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ 0 \\ -5 \end{pmatrix}$. **[2]**

(b) The cartesian equation of a line is $\frac{x}{3} = y - 2 = 2z + 6$. Express the equation of this line in vector form. **[3]**

2 The vector $\mathbf{p} = \begin{pmatrix} a^2 \\ a+1 \\ 12 \end{pmatrix}$, all of whose components are positive, is perpendicular to $\begin{pmatrix} 3 \\ 2 \\ -4 \end{pmatrix}$. Determine the exact value of a. **[4]**

3

(a) (i) Find $\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \times \begin{pmatrix} 3 \\ 5 \\ -2 \end{pmatrix}$. **[1]**

(ii) State a geometrical relationship between your answer to (i) and the two vectors, and (iii) verify the relationship. **[3]**

(b) Find the angle between the vectors $3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$ and $\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$. **[3]**

4

(a) Determine whether the point $(10, -1, 4)$ lies on the line $\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$. **[3]**

Vectors $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -4 \\ 3 \\ 12 \end{pmatrix}$.

(b) (i) Find, in degrees, the angle between **a** and **b**. **[3]** (ii) Find a vector perpendicular to both **a** and **b**. **[2]**

Question 1

(a) $\begin{pmatrix}2\\-1\\3\end{pmatrix}\times\begin{pmatrix}4\\0\\-5\end{pmatrix}=\begin{pmatrix}5\\22\\4\end{pmatrix}$

(b) $x=3\lambda$, $y=2+\lambda$, $z=-3+\frac{\lambda}{2}$: $\mathbf{r}=\begin{pmatrix}0\\2\\-3\end{pmatrix}+\lambda\begin{pmatrix}6\\2\\1\end{pmatrix}$ (taking $\lambda=2t$ gives direction $(6,2,1)$)

Question 2

$3a^2+2(a+1)-48=0\Rightarrow 3a^2+2a-46=0$, $a=\dfrac{-2\pm\sqrt{556}}{6}=\dfrac{-1\pm\sqrt{139}}{3}$. Components positive requires $a+1>0$, so $a=\dfrac{-1+\sqrt{139}}{3}=3.60$

Question 3

(a)(i) $\begin{pmatrix}1\\-1\\-1\end{pmatrix}$

(ii) It is perpendicular to both vectors. (iii) $(1,-1,-1)\cdot(1,2,-1)=1-2+1=0$ and $(1,-1,-1)\cdot(3,5,-2)=3-5+2=0$

(b) $\cos\theta=\dfrac{3-8-2}{\sqrt{14}\sqrt{21}}=\dfrac{-7}{\sqrt{294}}$, $\theta=114^\circ$

Question 4

(a) $x=10\Rightarrow\lambda=3$, then $y=2-3=-1$ (matches) but $z=-1+6=5\neq 4$. Not on the line

(b)(i) $\cos\theta=\dfrac{-4+6+24}{3\times 13}=\frac{2}{3}$, $\theta=48.2^\circ$

(ii) $\mathbf{a}\times\mathbf{b}=\begin{pmatrix}18\\-20\\11\end{pmatrix}$